CB.SC.U4CYS23015

Haemanth V J

Mid term lab evaluation:

1)

Dockerfile:



```
  GNU nano 8.7                                                    Dockerfile *
FROM python:3.11-slim

ENV ROLLNUMBER=CB.SC.U4CYS23015
ENV NAME="Haemanth VJ"

LABEL EMAIL="cb.sc.u4cys23015@cb.students.amrita.edu"
LABEL PRODUCTION_VERSION="1.0"

WORKDIR /midterm-exam

# install database and dependencies
RUN apt-get update && apt-get install -y mariadb-server mariadb-client && rm -rf /var/lib/apt/lists/*

COPY web/requirements.txt /midterm-exam/web/requirements.txt

RUN pip install -r /midterm-exam/web/requirements.txt

COPY database/ /midterm-exam/database/

COPY web/ /midterm-exam/web/

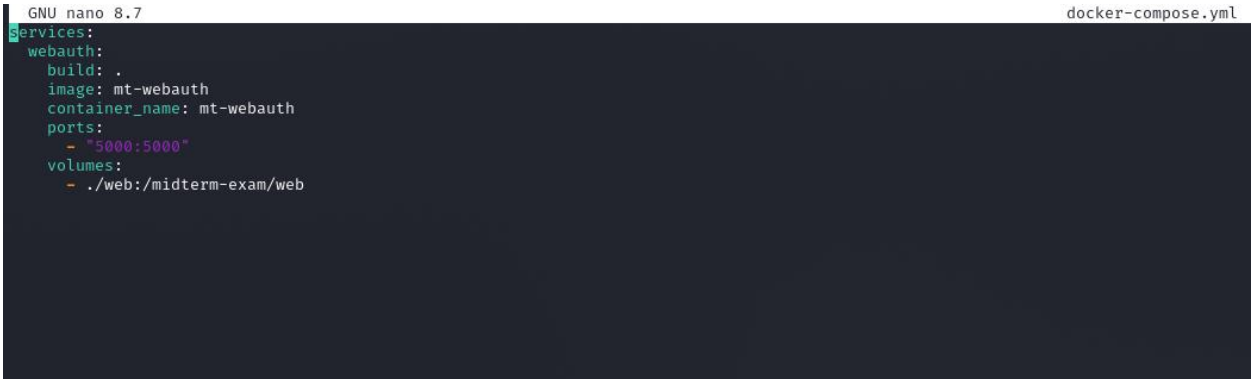
COPY entrypoint.sh /entrypoint.sh

RUN chmod +x /entrypoint.sh

EXPOSE 5000

CMD ["/entrypoint.sh"]
```

Docker-compose.yml



```
  GNU nano 8.7                                                  docker-compose.yml
services:
  webauth:
    build: .
    image: mt-webauth
    container_name: mt-webauth
    ports:
      - "5000:5000"
    volumes:
      - ./web:/midterm-exam/web
```

Entrypoint.sh



In web folder

App.py

Requirements.txt

```
 GNU nano 8.7                                                    app.py
import os
import mysql.connector
from flask import Flask, request, render_template

app = Flask(__name__)

rollnumber = os.environ.get("ROLLNUMBER", "N/A")
name = os.environ.get("NAME", "N/A")


def get_db():
    return mysql.connector.connect(
        host="127.0.0.1",
        user="midterm_user",
        password="midterm_pass",
        database="midterm_db"
    )


@app.route("/")
def index():
    return render_template("login.html", rollnumber=rollnumber, name=name)


@app.route("/login", methods=["POST"])
def login():
    username = request.form["username"]
    password = request.form["password"]

    db = get_db()
    cursor = db.cursor()
    cursor.execute(
        "SELECT * FROM users WHERE username = %s AND password = %s",
        (username, password)
    )
    user = cursor.fetchone()
    cursor.close()
    db.close()

    if user:
        return render_template("result.html", success=True,
                               message="Login successful. Welcome " + username + "!")
    else:
        return render_template("result.html", success=False,
                               message="Login failed. Wrong username or password.")


if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Requirements.txt

```
 GNU nano 8.7                                              requirements.txt
Flask
mysql-connector-python
```

A simple login page

Inside web folder /templates

Login.html

```
 GNU nano 8.7                                                    login.html
<!DOCTYPE html>
<html>
<head>
    <title>Login</title>
</head>
<body>
    <h1>Login Page</h1>
    <p>Roll Number: {{ rollnumber }} | Name: {{ name }}</p>

    <form action="/login" method="POST">
        <p>
            <label>Username:</label>
            <input type="text" name="username" required>
        </p>
        <p>
            <label>Password:</label>
            <input type="password" name="password" required>
        </p>
        <p>
            <button type="submit">Login</button>
        </p>
    </form>
</body>
</html>
```

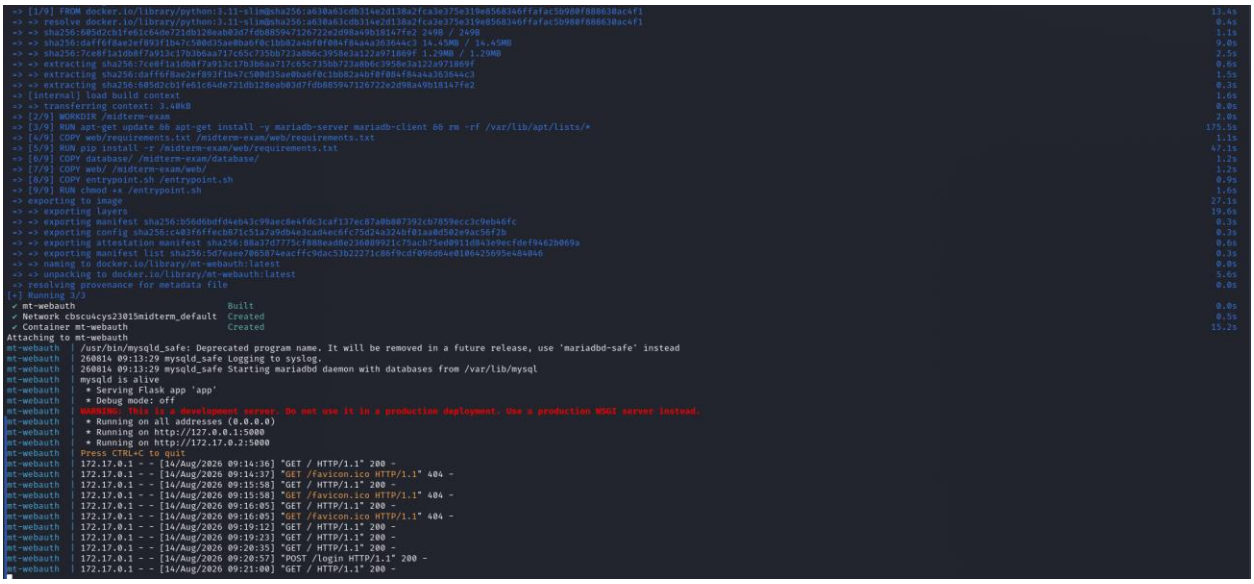
Result.html

```
 GNU nano 8.7                                                    result.html
<!DOCTYPE html>
<html>
<head>
    <title>Login Result</title>
</head>
<body>
    <h1>{{ 'Success' if success else 'Failed' }}</h1>
    <p>{{ message }}</p>
    <a href="/">Back to Login</a>
</body>
</html>
```

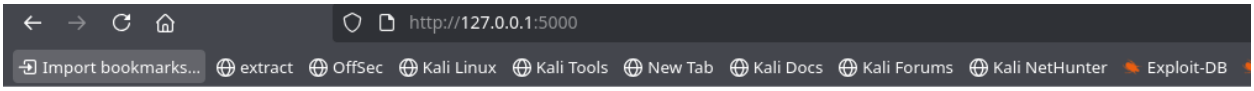
Inside database folder

Init.sql

```
 GNU nano 8.7                                                
```



Login page



Login Page

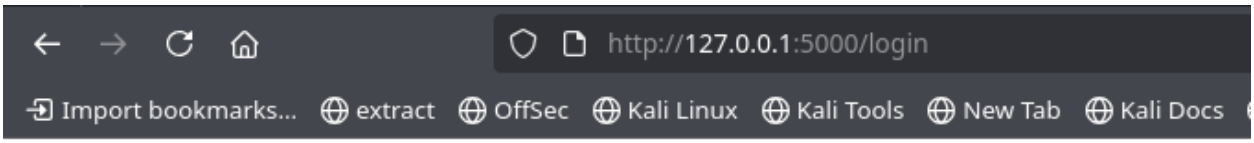
Roll Number: CB.SC.U4CYS23015 | Name: Haemanth VJ

Username:

Password:

Login

When we entry the wrong credentials

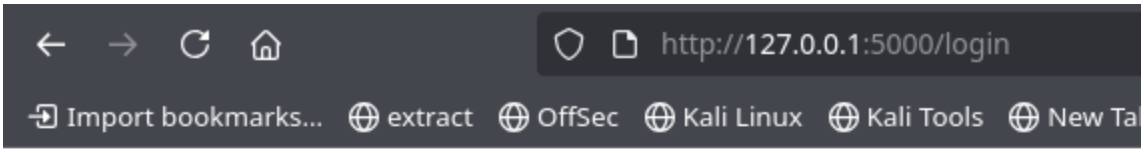


Failed

Login failed. Wrong username or password.

Back to Login

When we enter the right credentials



Success

Login successful. Welcome admin!

Back to Login

Creating tag for image

```
┌──(haems㉿haems)-[~/Music/CB.SC.U4CYS23015MIDTERM]
└─$ sudo docker tag mt-webauth haemanthvj/mt-webauth:1.0
```

Pushing the image in the docker hub

```
┌──(haems㉿haems)-[~/Music/CB.SC.U4CYS23015MIDTERM]
└─$ sudo docker push haemanthvj/mt-webauth:1.0
The push refers to repository [docker.io/haemanthvj/mt-webauth]
20340d661aee: Pushed
bda7559cf3b0: Pushed
d0bc93c6343f: Pushed
605d2cb1fe61: Pushed
9c423bb7f9bd: Pushed
9fc0df305b29: Pushed
26c307b5e35a: Pushed
e94999ddee7a: Pushed
889bb3f4c59f: Pushed
daff6f8ae2ef: Pushed
a540c25ad0c0: Pushed
7ce8f1a1db8f: Pushed
4af89f360f9d: Pushed
1.0: digest: sha256:42356e2a951bb05560f3da92ba9d533b23b6458bff098ead4a8d0de6d9382a04 size: 856

┌──(haems㉿haems)-[~/Music/CB.SC.U4CYS23015MIDTERM]
└─$
```

